

SIMATS ENGINEERING

ASSESSMENT TOOL2:INDUSTRYPROBLEMSOLVING TASK

CloudComputingandBigDataAnalytics–CSA1558

CO4: Analyze bigdatacharacteristics and challengesacross domains includingsecurity and application aspects.

Question Count: 5 | Total Marks: 40 | Duration: 60 Minutes

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Question 1: Retail Data Optimization Problem

A retail chain operates physical stores and an online platform and generates large volumes of sales, inventory, and customer-behavior data. The required solution must support real-time inventory tracking, demand forecasting, and personalized marketing recommendations.

1. Industry Context and Requirements

- ☐ Integrate sales, inventory, online activity, and customer behavior data.
- ☐ Process incoming data with low latency for real-time inventory visibility.
- ☐ Use historical data for demand forecasting.
- ☐ Analyze customer behavior for personalized marketing.
- ☐ Provide scalable storage and distributed processing for increasing data volumes.

2. Proposed Big Data Architecture

Retail Big Data Architecture

POS / Online Platform	Data Ingestion	Stream + Batch Processing	DataLake / Warehouse	Analytics & ML	Inventory / Marketing Dashboard
↓	↓	↓	↓	↓	

The architecture combines streaming and batch processing. Store POS systems and the online platform produce sales, inventory, and customer-event data. A distributed ingestion layer collects the data. Stream processing provides current inventory information, while batch processing analyzes historical data.

3. Tools, Technologies and Models

Component	Suitable technology / approach	Purpose
Data ingestion	Apache Kafka	Collect high-volume real-time events
Stream processing	Apache Spark Structured Streaming	Process inventory and customer events
Batch processing	Apache Spark	Process historical sales and behavior data
Storage	Distributed data lake / cloud object storage	Store raw and processed datasets
Analytics	Spark SQL / Big Data analytics	Analyze sales and inventory patterns
Demand forecasting	Time-series forecasting models	Predict future product demand
Personalization	Recommendation / customer-segmentation models	Generate targeted marketing recommendations

4. Working

1. POS and online systems generate sales, inventory, and customer-behavior events.
2. The ingestion layer collects the events continuously.
3. Stream processing updates inventory information with low latency.
4. Historical data is stored for large-scale batch analysis.
5. Forecasting models estimate future demand.
6. Customer analytics identifies preferences and behavior patterns.
7. Recommendation models generate personalized offers.
8. Dashboards provide inventory and marketing insights to business users.

5. Security and Risks

- ☐ Protect customer information using access control and encryption.
- ☐ Apply authentication and authorization to data and analytics services.
- ☐ Maintain data-quality checks to avoid incorrect inventory forecasts.
- ☐ Use backups and monitoring to reduce operational risks.

6. Expected Outcome

The solution provides real-time inventory visibility, supports data-driven demand forecasting, and enables personalized marketing while remaining scalable as the retail data volume grows.

Question2:SocialMediaFraudDetectionProblem

A social media platform processes billions of posts, comments, likes, and shares every day and needs to detect fake accounts and misinformation quickly. Therefore, the solution must support scalable stream processing and machine-learning-based detection.

1. Industry Context and Requirements

- ☐ Process very high volumes of user interactions.
- ☐ Detect suspicious account behavior with low latency.
- ☐ Identify potentially misleading or false information.
- ☐ Support continuous machine-learning analysis.
- ☐ Scale processing as platform activity increases.

2. Proposed Big Data Architecture

SocialMedia Fraud Detection Architecture

User Interactions	Kafka / Stream Ingestion	Stream Processing	Feature Extraction	MLDetection Models	Alerts / Moderation
↓	↓	↓	↓	↓	

3. Technologies and ML Approach

Component	Suitable technology / approach	Purpose
Event ingestion	Apache Kafka	Capture posts, likes, shares and other events
Stream processing Storage	Apache Spark Structured Streaming Distributed data lake	Process interaction streams continuously
Feature engineering	Distributed feature processing	Retain historical interactions and features
Fake-account detection	Classification / anomaly detection	Generate behavioral indicators
Misinformation detection	Text/content classification models	Identify suspicious account patterns
Monitoring	Real-time dashboards and alerts	Flag potentially misleading content
		Support rapid moderation decisions

4. Working

9. User interactions are continuously collected as events.
10. Stream processing analyzes events without waiting for a large batch.
11. Behavioral features such as unusual activity patterns are extracted.
12. Machine-learning models calculate fraud or misinformation risk.
13. High-risk accounts or content are sent to moderation/alert systems.
14. Historical data is used to retrain and improve the models.

5. Security and Risks

- ☐ Protect user data and restrict access to sensitive information.
- ☐ Monitor false positives and false negatives in automated detection.
- ☐ Prevent unauthorized manipulation of detection models or training data.
- ☐ Maintain audit logs for moderation decisions.

6. Expected Outcome

A streaming Big Data architecture enables rapid detection of suspicious behavior while distributed processing and machine learning allow the system to scale to very high interaction volumes.

Question3:GovernmentDataPlatformProblem

A government agency is building a national data platform combining census, economic, and geospatial datasets. Different departments need shared, integrated access while strict privacy and security requirements must be maintained.

1. Industry Context and Requirements

- ☐ Integrate datasets from multiple departments and domains.
- ☐ Provide interoperability between agencies.
- ☐ Maintain common data definitions and standards.
- ☐ Control access to sensitive government information.
- ☐ Ensure data quality, privacy, auditability, and accountability.

2. Proposed Governance and Data Management Architecture

Government Data Platform Architecture

Department Data Sources	Secure Ingestion	Integrated Data Lake	Metadata + Governance	Analytics / Shared APIs	Authorized Departments
↓	↓	↓	↓	↓	

3. Governance Strategy

Area	Strategy
Data standards	Use common schemas, formats, identifiers, and metadata standards
Data ownership	Assign clear data owners and custodians for each dataset
Access control	Use role-based access and least-privilege permissions
Data quality	Apply validation, deduplication, consistency and completeness checks
Metadata	Maintain a searchable catalog describing datasets and lineage
Privacy	Apply masking, anonymization/pseudonymization where appropriate

Security	Use encryption, authentication, monitoring and audit logs
Interoperability	Use standardized APIs and common data exchange formats

4. Working

15. Departments provide census, economic, and geospatial datasets.
16. Secure ingestion validates and transfers the data.
17. Data is integrated into a shared platform using common standards.
18. Metadata records describe sources, ownership, quality, and lineage.
19. Governance policies control who can access each dataset.
20. Authorized departments access integrated data through controlled services or APIs.
21. Auditing and monitoring track important access and data-management activities.

5. Risks and Controls

- ☐ Unauthorized access → strong authentication, authorization, and auditing.
- ☐ Privacy leakage → data minimization and appropriate masking/anonymization.
- ☐ Inconsistent datasets → common standards and quality validation.
- ☐ Data misuse → clear ownership, policies, access reviews, and audit trails.

6. Expected Outcome

The strategy creates a governed and interoperable national data platform where departments can share useful information while maintaining privacy, security, quality, and accountability.

Question 4: Banking Fraud Detection Problem

A bank processes millions of financial transactions every minute and requires real-time fraud detection using both historical and streaming transaction data. Low latency and high accuracy are critical.

1. Industry Context and Requirements

- ☐ Process millions of transactions with very low latency.
- ☐ Combine real-time transaction streams with historical customer behavior.
- ☐ Detect suspicious transactions before or immediately after authorization.
- ☐ Maintain high detection accuracy.
- ☐ Scale processing during high transaction periods.

2. Proposed Big Data Pipeline

Banking Fraud Detection Pipeline

Transaction Stream	Kafka / Ingestion	Stream Processing	FeatureStore+ Historical Data	Fraud ML Model	Risk Score / Alert
↓	↓	↓	↓	↓	

3. Technologies and Algorithms

Component	Suitable technology / approach	Purpose
Streaming ingestion	Apache Kafka	Receive transaction events continuously
Stream processing	Apache Spark Structured Streaming	Compute real-time transaction features
Historical storage	Distributed data lake / warehouse	Store previous transactions and customer history
Feature engineering	Distributed processing	Calculate behavioral and transaction indicators
Fraud algorithm	Classification and anomaly detection	Identify suspicious transaction patterns

Decision layer	Risk scoring / rules	Combine model score with business rules
Alerting	Real-time alert service	Notify fraud systems or investigators

4. Working

22. A transaction is received by the streaming ingestion layer.
23. The transaction is processed immediately.
24. Real-time features are combined with relevant historical customer behavior.
25. The fraud model calculates a risk score.
26. Business rules and risk thresholds are applied.
27. High-risk transactions generate alerts or additional verification.
28. Historical outcomes are retained for model evaluation and improvement.

5. Security and Risks

- ☐ Encrypt sensitive transaction data in transit and at rest.
- ☐ Use strict access controls for financial data.
- ☐ Maintain audit logs for fraud decisions and investigator actions.
- ☐ Monitor model performance to reduce false positives and false negatives.
- ☐ Design for fault tolerance so fraud detection continues during infrastructure failures.

6. Expected Outcome

The pipeline combines real-time streaming with historical analytics to provide fast and accurate fraud detection while supporting large transaction volumes.

Question 5: Agriculture Smart Farming Problem

A smart farming system collects soil, weather, and crop data from sensors and drones. Farmers require insights to improve crop yield and resource usage, while data is generated from multiple distributed sources.

1. Industry Context and Requirements

- ☐ Collect data from soil sensors, weather systems, drones, and crop observations.
- ☐ Handle distributed and continuous data sources.
- ☐ Analyze current field conditions.
- ☐ Predict crop and resource requirements.
- ☐ Provide practical recommendations for irrigation, fertilization, and crop management.

2. Proposed Big Data Architecture

Precision Agriculture Big Data Architecture

Sensors / Drones / Weather	Data Ingestion	Stream + Batch Processing	Data Lake	Analytics / ML	Farmer Dashboard / Recommendations
↓	↓	↓	↓	↓	

3. Technologies and Analytics

Component	Suitable technology / approach	Purpose
Data collection	IoT sensor gateways / APIs	Collect field and weather data
Streaming	Apache Kafka	Receive continuous sensor events
Processing	Apache Spark	Analyze large historical and streaming datasets
Storage	Distributed data lake	Store sensor, crop, weather and drone

		datasets
Crop analytics	Statistical / ML models	Identify relationships between conditions and crop outcomes
Yield prediction	Regression / forecasting models	Estimate expected crop yield
Resource optimization	Rule-based or ML recommendations	Improve water and fertilizer usage
Visualization	Farmer dashboard	Present field insights and recommendations

4. Working

- 29. Sensors, drones, and weather sources collect field information.
- 30. The ingestion layer transfers data to the Big Data platform.
- 31. Streaming processing provides near real-time field information.
- 32. Historical data is processed to identify long-term patterns.
- 33. Analytics models estimate crop condition and yield.
- 34. The system identifies irrigation, fertilizer, or other resource requirements.
- 35. Recommendations are displayed to farmers through a dashboard.

5. Security and Risks

- ☐ Protect farm and operational data using authentication and access controls.
- ☐ Validate sensor data to detect faulty or abnormal readings.
- ☐ Use backups and fault-tolerant storage for important datasets.
- ☐ Monitor distributed devices and data sources for failures.

6. Expected Outcome

The solution converts distributed sensor, weather, crop, and drone data into actionable insights that can help farmers improve yield and use water, fertilizer, and other resources more efficiently.

Rubric Alignment – Level 4 Target

The answers are organized to address the assessment rubric for the Industry Problem Solving Task:

Rubric Criterion	How the answers address it
Understanding of Industry Context (20%)	Each answer identifies the real-world problem, data sources, and operational requirements.
Application of Big Data Concepts (25%)	Each solution applies relevant ingestion, distributed processing, storage, analytics, and ML concepts.
Solution Design (25%)	Each problem includes a practical and scalable architecture with a clear data flow.
Security / Risk Awareness (15%)	Each solution discusses security, privacy, data quality, reliability, or operational risks relevant to the problem.
Clarity and Professionalism (15%)	Answers use consistent headings, tables, numbered steps, diagrams, and concise technical explanations.

Note: The technology choices above are proposed solutions based on the requirements stated in the uploaded assessment. They are presented as suitable design options rather than as technologies explicitly mandated by the question.